Simulating earthquake waves 8

Read the diary entry and follow the instructions below.

Dear Diary
Today Ms Smart explained and demonstrated the difference between **primary** and **secondary** earthquake waves. She said primary waves were **compression waves**, while secondary waves were **transverse waves**. At first I did not understand, but then Ms Smart showed us an activity that clearly demonstrated the difference between the two types of waves.

We were asked to select a partner; Ms Smart was my partner! In pairs, we stretched out a slinky spring about 5 metres across the floor. I held one end, while Ms Smart held the other end and gathered about 15 coils and let them go. She said that these were what compression waves looked like, similar to primary waves that travelled through rock. We also noticed that the primary waves travelled along the slinky faster than the secondary waves, reaching my hand sooner.

We stretched the slinky again about 5 metres. This time Ms Smart asked me to move my end of the slinky side to side, about 30 cm to each side, in a very quick movement. We saw a sideways pulse travelling down the slinky to the other end. Ms Smart said these were exactly what secondary earthquake waves were like.

When I got home, I showed Mum the different waves on my slinky, but she just said, "How interesting, now go and do your homework." Parents! Can't teach them anything!

Title:

Aim: _____

_____

Equipment:

Method: _____

_____

_____

_____

_____

_____

_____

_____

_____

Results:

Conclusion: _____

_____

_____

_____

1 Name the text type that the diary entry is written in? _____

2 Change the diary entry into an experimental record text type. Rewrite the text in the space provided above and complete it by suggesting possible results and conclusion.

3 State another name for:

a a primary wave. _____

b a secondary wave. _____

4 **a** Imagine a small rock in the earth's crust. Describe how the rock would move if a primary wave passed through it.

_____

_____

b Think of the same small rock again and describe how it would move if a secondary wave passed through it.

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_____

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